

Functionally Decoupled Visual Intent Learning under Supervisory and Semantic Ambiguity

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Abstract

Visual intention understanding aims to infer the latent purpose conveyed by an image, and is fundamentally more challenging than conventional image classification due to its abstract and subjective nature. A central difficulty of this task lies in two complementary forms of ambiguity: supervisory ambiguity, caused by subjective annotation over plausible intentions, and semantic ambiguity, caused by semantically adjacent and overlapping intents. In this work, we propose FDIL (**F**unctionally **D**ecoupled **I**ntent **L**earning), a framework that treats these two sources of ambiguity through complementary roles. FDIL first introduces a semantic prior built from textual cues to provide structured candidate semantics, and then learns image-aware intent prediction under uncertainty-guided teacher supervision. Concretely, we develop Semantic Local Reranking and Calibration (SLR-C) to form the semantic prior, and Uncertainty-guided Text Distillation (UTD) to transform annotator disagreement into an adaptive training signal. By combining semantic structure with uncertainty-aware learning, FDIL improves both the stability of supervision and the discrimination of semantically similar intents. **On the Intentonomy dataset, FDIL outperforms representative recent state-of-the-art methods by at least +3.4 in average F1, +5.7 in hard-subset F1, and +3.4 in threshold-free mAP. Extensive experiments demonstrate consistent gains in both hard-sample robustness and overall multi-label prediction performance.**

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Keywords: Visual Intent recognition, Image Understanding, Multi-label Image Classification

1. Introduction

With the rapid proliferation of social media, images are increasingly used as a medium for users to share daily experiences, emotions, and personal viewpoints. Visual intention recognition aims to infer the latent purpose conveyed by an image, and has become an important problem in human-centered multimedia analysis [1, 2]. Accurately recognizing these intentions is valuable for a variety of downstream applications, including user interest understanding, psychological and behavioral profiling, broader social media content analysis, *etc.*

Compared with conventional image classification, visual intention recognition is fundamentally more challenging. In traditional classification tasks, the goal is to assign images to relatively objective semantic categories, such as objects, scenes, or actions. In contrast, intention recognition requires mapping visual evidence to abstract and subjective intent labels. These labels are rarely determined by local visual cues alone; instead, they often emerge from a holistic interpretation of scene context, human activities, object configurations, social interactions, and commonsense knowledge. Therefore, the main difficulty of this task lies not merely in visual representation learning, but in establishing a reliable bridge between concrete image content and abstract intention semantics.

A central challenge of this task is its inherent ambiguity, which manifests in two distinct forms, as shown in Figure 1 (a). The first is semantic ambiguity. Intent labels are often semantically adjacent, partially overlapping, or hierarchically correlated. Different labels may correspond to visually similar patterns, while the same intent may be expressed through diverse appearances. This leads to high inter-class similarity and substantial intra-class variation, making fine-grained discrimination particularly difficult. The second is supervisory ambiguity. Since intention recognition is intrinsically subjective, different audiences may infer different yet plausible intentions from the same image. As a result, the supervisory signal itself may become unstable, especially

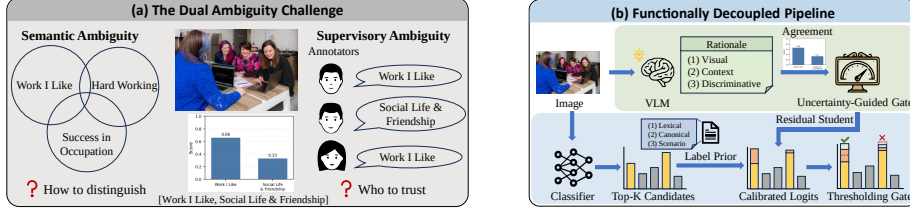


Figure 1: **Functionally decoupled ambiguity modeling in FDIL.** (a) Visual intent recognition suffers from semantic ambiguity among overlapping intents and supervisory ambiguity from annotator disagreement. (b) FDIL separates semantic refinement via Semantic Local Reranking and Calibration (SLR-C) from uncertainty-aware semantic supervision via Uncertainty-guided Text Distillation (UTD).

for difficult samples with severe disagreement. Under such circumstances, directly optimizing against hard labels can force the model to fit oversimplified or unreliable targets, thereby harming the robustness of the learned decision boundaries. Recent studies have increasingly argued that such disagreement should not be treated as mere annotation noise, but as meaningful uncertainty that reflects the structure of subjective interpretation itself [3, 4].

Existing studies have attempted to alleviate these difficulties from different perspectives. Prototype-based methods, *e.g.* PIP-Net [5], introduce representative anchors to reduce feature-space confusion. Structure-aware methods, including CPAD [6] and HLEG [7], exploit hierarchical label relationships to enhance multi-level semantic modeling and strengthen supervision. From the perspective of label quality, LabCR [8] explicitly models label shifting and label blemish to better cope with imperfect annotation distributions. More recently, knowledge-enhanced approaches have recognized that visual evidence alone is often insufficient for subjective intention understanding. IntCLIP [9] incorporates the vision-language alignment knowledge of CLIP into the task, while IntentMLM [10] further leverages multimodal large models, caption generation, and retrieval to perform more holistic intention perception. **This line of work reflects a broader recent shift toward large vision-language and multi-modal large language models (MLLMs) as sources of external semantic knowledge for subjective image understanding. However, our zero-shot experiments show that prompting a general-purpose MLLM directly remains below a task-trained model on**

this benchmark. Such models supply useful intent priors but do not provide the dataset-specific calibration and uncertainty-aware supervision that subjective intent recognition requires. Our purpose is therefore not to replace these models, but to convert their semantic knowledge into a structured prior and an uncertainty-gated training signal that explicitly target the two ambiguities below.

Although these methods have yielded encouraging progress, they typically treat one perspective of ambiguity as dominant. In our view, the core difficulty of subjective visual intention recognition is not any single ambiguity alone, but the coexistence of supervisory ambiguity and semantic ambiguity, which interfere with different functional components of the system. The former primarily corrupts supervision by compressing diverse subjective interpretations into rigid targets, whereas the latter primarily distorts local decision geometry by making neighboring labels difficult to separate. Treating these two failure modes within one mechanism may therefore be suboptimal, as it is difficult to simultaneously achieve robust learning on disagreement-prone samples and precise discrimination among closely related intents.

Motivated by this observation, we propose the **FDIL** (**F**unctionally **D**ecoupled **I**ntent **L**earning), a functionally decoupled framework for subjective intent recognition, as illustrated in Figure 1(b), where fixed semantic priors provide structured candidate semantics, uncertainty-aware text teachers stabilize ambiguous supervision, and a residual student learns image-aware intent prediction on top of the semantic prior. We decouple ambiguity handling into two complementary functions. First, Semantic Local Reranking and Calibration (SLR-C) provides ambiguity-aware semantic refinement: heterogeneous lexical, canonical, and scenario priors define a semantic prior that supplies structured candidate semantics. Second, Uncertainty-guided Text Distillation (UTD) provides ambiguity-aware semantic supervision: a text-only teacher distilled from rationale embeddings regularizes the visual model according to annotator agreement, improving robustness under subjective supervision and supporting image-aware prediction on top of the semantic prior. Together, these two functions improve both robustness and discriminability without introducing high architectural complexity.

Extensive experiments on the Intentionomy dataset show that our FDIL consistently outperforms strong existing baselines and recent state-of-the-art methods, yielding im-

provements in both hard-sample robustness and overall multi-label prediction quality.

The main contributions of this work are summarized as follows:

- We reformulate visual intention recognition as a *functionally decoupled* ambiguity problem, **treating supervisory ambiguity and semantic ambiguity as two separately justified functional roles rather than a single undifferentiated mechanism. This decomposition outperforms a strong unified alternative while keeping each function individually interpretable, separately measurable, and independently deployable.**
- We propose Semantic Local Reranking and Calibration (SLR-C) together with residual semantic refinement, where heterogeneous textual priors first define a semantic prior with structured candidate semantics and a lightweight residual student further adapts prediction to image evidence.
- We propose Uncertainty-guided Text Distillation (UTD), which leverages annotator disagreement as an uncertainty signal to adaptively introduce semantic guidance, improving robustness under ambiguous supervision.

2. Related Work

Early visual intent studies target specific domains such as persuasive or political imagery [11, 12, 13], while the Intentionomy benchmark [1] enables general-purpose intent understanding in unconstrained social media. Recent Intentionomy methods address the task from complementary angles: PIP-Net [5] uses prototype-based representations for interpretable intent perception; CPAD [6] and HLEG [7] exploit cross-modality or hierarchical label structures; LabCR [8] models label calibration and relation preservation; and IntCLIP [9] and IntentMLM [10] introduce CLIP/MLLM-based semantic knowledge. Robust-loss methods such as Focal Loss [14], OHEM [15], collaborative refining [16], and ASL [17] reduce the effect of difficult, noisy, or imbalanced samples through reweighting or selective optimization; however, they treat ambiguity mainly as an optimization issue rather than as a subjective semantic signal. Disagreement-aware learning instead preserves label diversity: Peterson et al. [3]

train on human label distributions, Davani et al. [4] model annotator-specific labels, DisCo [18] jointly models item- and annotator-level distributions, and CrowdOpinion [19] recovers minority opinions. In contrast, FDIL explicitly treats annotator disagreement as an uncertainty signal and adaptively regulates semantic guidance according to supervision ambiguity, yielding more robust learning under subjective supervision.

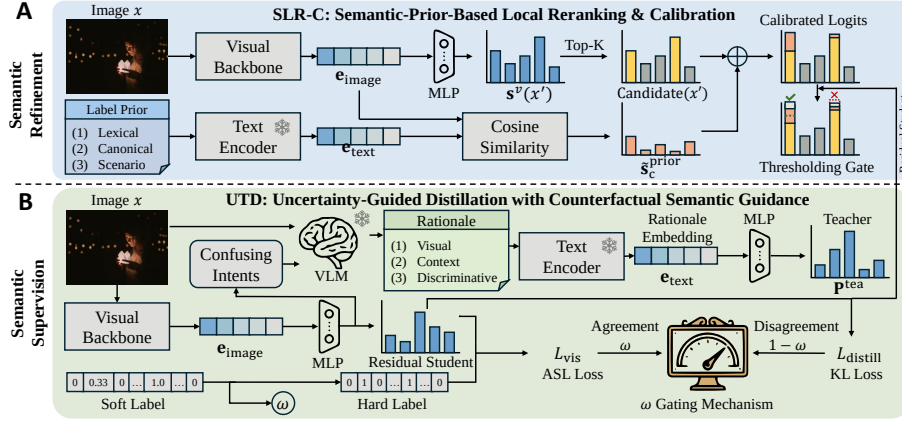


Figure 2: Overview of FDIL. SLR-C constructs a textual semantic prior and performs local reranking plus residual refinement. UTD uses a rationale-based text teacher and an agreement-aware gate to stabilize learning under ambiguous supervision.

3. Method

3.1. Overview

To address the dual challenges of supervisory ambiguity and semantic ambiguity in visual intention recognition, we propose the **FDIL**, a functionally decoupled framework where fixed semantic priors provide structured candidate semantics, uncertainty-aware text teachers stabilize ambiguous supervision, and a residual student learns image-aware intent prediction on top of the semantic prior. Instead of forcing one unified mechanism to handle both supervisory ambiguity and semantic ambiguity, we assign them to two complementary functions.

FDIL consists of two key components, as shown in Figure 2:

- **Semantic Local Reranking and Calibration (SLR-C)**, which constructs a semantic prior from heterogeneous textual cues and supports image-aware refinement through a residual student;
- **Uncertainty-guided Text Distillation (UTD)**, which introduces adaptive semantic guidance to improve robustness under ambiguous supervision.

Given an input image, the model first produces visual intent scores using a strong visual backbone. SLR-C then provides structured candidate semantics by locally reranking intent candidates and forming a semantic prior. UTD supplies uncertainty-aware semantic supervision through a rationale-based text teacher and can be reused to regularize both the visual student and the residual refinement module built on top of the semantic prior.

Table 1: **Key notation used in the method.**

Symbol	Meaning
x	input image
$\mathcal{Y}, C$	intent label space and number of classes
$\mathbf{y}^{\text{soft}}, \mathbf{y}^{\text{hard}}$	soft annotation / binarized hard labels
$\omega \in (0, 1]$	sample-level annotator agreement score
$\phi = f_{\text{img}}(x)$	frozen visual image feature
$\bar{s}_c^{\text{prior}}$	normalized heterogeneous semantic prior
$C_K(x), K$	Top- K candidate set and its size
s_c^{full}	final refined score for class c
t_c	class-wise decision threshold (validation-only)
$\mathbf{p}^{\text{tea}}, \tau$	text-teacher distribution and temperature
$g = 1 - \omega$	teacher-branch gate weight
$\mathcal{L}_{\text{total}}$	total training loss

3.2. Problem Formulation

Given an input image x , the goal is to predict a set of intent labels from a predefined label space $\mathcal{Y} = \{1, \dots, C\}$, where C is the total number of intent categories. Due to the

subjective nature of visual intention recognition, annotations are provided as soft label vectors $\mathbf{y}^{\text{soft}} \in \{0, 0.33, 0.66, 1\}^C$, reflecting annotator agreement.

Existing approaches typically follow either hard binarization or soft regression paradigms, both of which have limitations under ambiguous supervision. To address this, we adopt a **decoupled utilization** of annotation signals.

Specifically, we derive binarized labels $\mathbf{y}^{\text{hard}}$ for classification supervision, while extracting an image-level agreement score $\omega \in (0, 1]$ from $\mathbf{y}^{\text{soft}}$. Here, ω reflects annotator consensus, where higher values indicate more reliable supervision. The binarized labels are computed as:

$$y_c^{\text{hard}} = \begin{cases} 1, & \text{if } y_c^{\text{soft}} > 0 \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

In practice, an image may contain multiple positive intents with different agreement levels. We therefore adopt a conservative pooling rule and define the sample-level agreement as the minimum soft agreement among all positive labels:

$$\omega = \min_{c: y_c^{\text{hard}}=1} y_c^{\text{soft}}. \quad (2)$$

This choice lets the hardest positive label determine how much the sample should trust direct supervision.

3.3. Semantic Local Reranking and Calibration (SLR-C)

Semantic ambiguity mainly manifests as local confusion among visually plausible but semantically adjacent intents. To address this, we first build a semantic refinement module that provides structured candidate semantics before turning to ambiguity-aware supervision.

3.3.1. Semantic Prior Construction

Intent semantics are inherently multi-granular and abstract, making a single textual representation insufficient. We therefore construct a heterogeneous semantic prior for each intent by combining three complementary textual sources: lexical intent phrases, canonical intent descriptions, and descriptive visual scenarios. The lexical source provides concise intent name cues, the canonical source provides semantic definitions, and the scenario source grounds abstract intents into plausible visual situations.

Example. For the intent *SocialLifeFriendship*, the **lexical** phrase is “*Social Life and Friendship*”, and the canonical description is: “*Being part of a social group. Having people to do things with. Having close friends, others to rely on. Making friends, drawing others near.*” This intent is further expanded into multiple scenario archetypes with distinct visual realizations; **for *SocialLifeFriendship*, scenarios include friends hiking together, close friends talking on a couch, a group dinner, or volunteers gardening together.**

The scenarios are not hand-picked but produced by a fixed generation protocol applied uniformly to every intent class (template released with our artifacts): the vision-language model enumerates visually distinct realizations along five axes (subjects/objects, actions, settings, atmosphere, and photographic style), with at least three settings per intent so as to span intra-class variation rather than a single canonical view. Inclusion favors representativeness over exhaustive coverage; outside the generated scenarios the prior degrades gracefully toward the lexical and canonical sources, the base logits, and the residual student.

Given an image x , we score each intent by the CLIP-scaled cosine similarity between the image and text embeddings:

$$s_c^{\text{prior}} = \text{sim}(f_{\text{img}}(x), \mathbf{e}_c) = \gamma \cdot \frac{f_{\text{img}}(x)^\top \mathbf{e}_c}{\|f_{\text{img}}(x)\|_2 \|\mathbf{e}_c\|_2}, \quad (3)$$

where γ is the CLIP logit scale. Each source $r \in \{\text{lex}, \text{can}, \text{scen}\}$ is normalized independently across the C classes and the three are averaged into the prior:

$$\tilde{s}_c^r = \frac{s_c^r - \mu^r(x)}{\sigma^r(x) + \epsilon}, \quad \tilde{s}_c^{\text{prior}} = \frac{1}{3} \sum_r \tilde{s}_c^r, \quad (4)$$

where $\mu^r(x)$ and $\sigma^r(x)$ are the per-sample mean and standard deviation over the C intent classes and ϵ ensures numerical stability.

3.3.2. Semantic Local Reranking

In the refinement module, we start from base visual logits s^b produced by a frozen visual student. We first select a compact candidate set as:

$$C_K(x) = \text{TopK}(s^b, K). \quad (5)$$

The base logits are then locally adjusted with the normalized prior:

$$\tilde{s}_c = \begin{cases} s_c^b + \alpha \cdot \tilde{s}_c^{\text{prior}}, & c \in C_K(x), \\ s_c^b, & \text{otherwise.} \end{cases} \quad (6)$$

Here α controls the strength of semantic reranking. In our implementation, α is fixed to 0.3.

This localized reranking produces a semantic prior that narrows the candidate set and sharpens the local score geometry without introducing an additional trainable prior module.

3.3.3. Residual Refinement on the Semantic Prior

Fixed reranking alone is not sufficient, because the heterogeneous prior may still fall short of the image-specific evidence. We therefore train a lightweight residual student on top of the semantic prior:

$$s_c^{\text{full}} = \tilde{s}_c + h_{\text{res}}(f_{\text{img}}(x))_c, \quad (7)$$

where $h_{\text{res}}(\cdot)$ is a residual MLP that predicts class-wise residual intent scores from the image feature. In practice, the SLR-C logits $\tilde{s}_c$ are first computed and frozen, and the residual module is then optimized separately.

In the final full model, this residual module is not trained in isolation: it can also be regularized by the uncertainty-aware text teacher introduced next. Thus, SLR-C provides the structured candidate semantics first, and UTD later supplies the ambiguity-aware supervision used to stabilize image-aware prediction on top of the semantic prior.

3.3.4. Class-wise Calibration

Motivated by Guo et al. [20], to account for class imbalance and varying score distributions, we adopt class-wise thresholds:

$$\hat{y}_c = \mathbb{I}[\sigma(s_c^{\text{full}}) > t_c], \quad (8)$$

where t_c is determined on the validation set. Concretely, each threshold is selected independently by maximizing the class-wise validation F1 score:

$$t_c^* = \arg \max_{t \in \mathcal{T}} \text{F1}_c^{\text{val}}(t), \quad (9)$$

where $\mathcal{T}$ is a fixed threshold grid. This design lets each class adapt its decision boundary to its own score distribution and frequency level, instead of sharing a single global threshold.

3.4. Uncertainty-guided Text Distillation (UTD)

Supervisory ambiguity arises when multiple plausible intents coexist for a given image. Direct optimization against hard labels may lead to overconfident predictions and unstable decision boundaries. To alleviate this issue, we introduce semantic guidance from text space and adapt its influence based on the uncertainty of supervision.

3.4.1. Text-derived Teacher

For each training image, we construct structured textual descriptions that capture semantic cues relevant to intention understanding. These descriptions are generated offline by a vision-language model, guided by both the image content and intent semantics.

To enhance discriminative capability under semantic ambiguity, we further incorporate contrastive cues by identifying visually plausible but incorrect intents predicted by the current model. Specifically, we select high-confidence negative intents as confusion candidates and use them to guide the generation of discriminative textual descriptions. This encourages the resulting text to emphasize subtle distinctions among semantically adjacent intents.

The generated text aims to summarize three aspects: (1) observable visual elements, (2) contextual interpretation beyond direct perception, and (3) discriminative cues for distinguishing confusing intents.

The resulting text $T(x)$ is encoded into a dense embedding space using a frozen text encoder:

$$\mathbf{e}_{\text{text}} = f_{\text{text}}(T(x)). \quad (10)$$

A lightweight classifier then maps the embedding into a multi-label teacher distribution:

$$\mathbf{p}^{\text{tea}} = \sigma(g_{\text{tea}}(\mathbf{e}_{\text{text}})/\tau), \quad (11)$$

where $\sigma(\cdot)$ is the sigmoid function and τ is the temperature parameter.

This design introduces contrastive semantic signals that improve the teacher’s ability to distinguish semantically similar intents, while remaining separate from the semantic prior itself. In the full FDIL pipeline, the same teacher can further regularize the residual refinement module built on top of SLR-C.

3.4.2. Uncertainty-guided Distillation

The visual backbone produces logits $\mathbf{s}^v$ for image x . We adopt the Asymmetric Loss [17] for classification:

$$\mathcal{L}_{\text{vis}} = \frac{1}{C} \sum_{c=1}^C \ell_{\text{ASL}}(s_c^v, y_c^{\text{hard}}). \quad (12)$$

To incorporate semantic guidance, we introduce an uncertainty-guided Bernoulli distillation loss:

$$\mathcal{L}_{\text{distill}} = \sum_{c=1}^C \text{KL}(\text{Bern}(p_c^{\text{tea}}) \parallel \text{Bern}(\sigma(s_c^v/\tau))), \quad (13)$$

which is applied independently to each label. We use Bernoulli-style distillation rather than softmax distillation because intent recognition is a multi-label problem: each intent is modeled as an independent Bernoulli variable rather than as a mutually exclusive class on a probability simplex. This formulation preserves per-class uncertainty and avoids imposing artificial competition among labels.

Rather than mixing the two objectives with a fixed global coefficient, we use a gate derived from the agreement score. For the current sample, the teacher branch weight is defined as $g = 1 - \omega$, so low-agreement samples receive stronger semantic guidance, while high-agreement samples remain dominated by direct visual supervision.

Using g , the final loss is defined as:

$$\mathcal{L}_{\text{total}} = (1 - g)\mathcal{L}_{\text{vis}} + g\mathcal{L}_{\text{distill}} = \omega\mathcal{L}_{\text{vis}} + \lambda(1 - \omega)\mathcal{L}_{\text{distill}}. \quad (14)$$

This gated formulation allows the model to rely more on visual supervision when annotations are reliable, and to leverage semantic guidance when supervision is uncertain. In other words, UTD stabilizes ambiguous supervision through an uncertainty-aware text teacher rather than by modifying the semantic prior itself.

3.5. Functionally Decoupled Ambiguity Modeling

We interpret visual intention recognition as a functionally decoupled ambiguity problem: semantic ambiguity primarily distorts local candidate semantics, while supervisory ambiguity primarily destabilizes semantic supervision.

A candidate-recall diagnostic supports this local view. The baseline Top-10 candidates already cover 83.9% of positive labels, and 67.8% of its false negatives remain recoverable within the same set on Intentionomy. Thus, the main bottleneck is often ranking and calibration among retrieved neighboring intents rather than failure to retrieve the correct intent.

A short risk-decomposition argument explains why two separately parameterized functions are preferable. Let $\phi = f_{\text{img}}(X)$ be the frozen image feature, $\tilde{y}_c$ the noisy annotation, and $\eta_c(x) = \Pr(Y_c=1 \mid X=x)$ the Bayes posterior. For class c , the per-sample excess risk splits into a label-side and a feature-side term:

$$\mathbb{E}[\ell(g_c(\phi), \tilde{y}_c)] \approx \underbrace{\mathbb{E}_{\mathcal{X}}[\mathcal{H}(\eta_c(X))]}_{\text{(S) supervisory}} + \underbrace{\mathbb{E}_{\mathcal{X}}[\eta_c(X) \text{ aliasing}_c(\phi)]}_{\text{(F) semantic}} + \text{est.}, \quad (15)$$

where $\mathcal{H}$ is the binary entropy. Term (S) depends only on $\Pr(Y \mid X)$, is large where annotators disagree, and is irreducible by any function of ϕ ; term (F) measures the class- c evidence in ϕ that is inseparable from a neighbor c' and is reduced only by restructuring the feature-side geometry. Because the two terms are functionals of different objects, a single mixed-target mechanism must trade one against the other, whereas a decoupled realization need not. The two FDIL modules instantiate exactly these corrections: SLR-C attacks term (F) by contracting the candidate set and adding image-specific residual directions $h_{\text{res}}(\phi)$ without touching the supervision target, while UTD attacks term (S) by down-weighting the direct loss through the agreement gate $g = 1 - \omega$ where $\mathcal{H}(\eta_c(X))$ is high. This design handles ambiguity without heavy architectural complexity.

4. Experiments

4.1. Experimental Setup

4.1.1. Dataset and Evaluation Metrics

We evaluate our method on the Intentionomy dataset [1], a multi-label benchmark for visual intention recognition in social media scenarios. Following the standard split used in our experiments, the dataset contains 12,739 training images, 498 validation images, and 1,216 test images. Intentionomy additionally provides soft agreement annotations with values in $\{0, 0.33, 0.66, 1\}$ on the training set, which directly reflect the degree of annotator consensus and serve as the uncertainty signal ω in our framework.

We use Macro F1, Micro F1, Samples F1 as the evaluation metrics. To be consistent with recent Intentionomy works, we additionally report *Avg.*, which is the average of three F1 scores. **Unless otherwise stated, all reported numbers are percentages and all results are computed on the Intentionomy test split.**

To better understand the effectiveness of our method, we further conduct experiments on subsets of the intent classes. We follow the work of Jia et al. [1] and break down intent classes according to content difficulty and dependency. Difficulty classes are measured by how much a standard ResNet-50 model outperforms the random results. Content classes are divided into object-dependent (*Object*-classes), context-dependent (*Context*-classes), and *Others*, which depend on both foreground and background information. We emphasize the *Hard* score throughout the analysis because it specifically measures performance on ambiguity-prone intent categories, where semantic overlap and subjective disagreement are most severe. Improvements on the hard subset **therefore provide** stronger evidence that a method is resolving the core difficulty of intention understanding rather than merely exploiting easy classes.

4.1.2. Implementation Details

Our default visual backbone is frozen CLIP ViT-L/14 [33]. The visual student is implemented as a lightweight residual MLP trained on top of a fixed SLR-C prior. In SLR-C, priors are constructed from three sources, namely lexical intent phrases, canonical intent descriptions, and Gemini 3.1 Pro-generated scenario descriptions [34]

Table 2: Main comparison on Intentonomy test split. Avg. is the mean of Macro/Micro/Samples F1. FDIL rows report mean $\pm$ std over three seeds. [†]Quoted from IntentMLM [10].

Method	Publication	Macro F1	Micro F1	Samples F1	Avg.
Random	-	6.94	7.18	7.10	7.07
Visual [21]	CVPR 2016	22.77	30.23	28.45	27.15
Multi-label classification methods					
HiMulConE [22]	CVPR 2022	28.43	41.50	42.15	37.36
MultiGuard [23]	NeurIPS 2022	26.87	38.67	40.06	35.20
SST [24]	AAAI 2022	27.33	42.21	42.62	37.39
Query2Label [25]	-	32.12	44.64	45.15	40.64
DualCoOp [26]	NeurIPS 2022	31.06	44.42	40.40	38.63
Label noise methods					
Cleannet [27]	CVPR 2018	25.89	42.15	40.97	36.34
LS [28]	ICML 2020	24.32	42.77	43.27	36.79
Dividemix [29]	ICLR 2020	30.11	40.23	41.56	37.30
Intent perception methods					
Jia [1]	CVPR 2021	23.98	31.28	31.39	28.88
CPAD [6]	TIP 2023	27.39	40.98	41.12	36.50
HLEG [7]	TAC 2023	32.77	44.69	45.54	41.00
HLEG (CLIP ViT-L/14) [7]	TAC 2023	48.17	57.29	56.81	54.09
PIP-Net [5]	TMM 2023	32.57	45.94	47.05	41.85
LabCR [8]	TIP 2024	34.63	48.51	48.05	43.73
LabCR (CLIP ViT-L/14) [8]	TIP 2024	48.04	57.13	56.65	53.94
IntCLIP [9]	ECCV 2024	40.15	53.54	51.01	48.23
IntCLIP (CLIP ViT-L/14) [9]	ECCV 2024	47.28	56.18	54.86	52.77
IntentMLM [10]	PR 2026	43.05	54.77	56.75	51.52
Zero-shot VLM					
Qwen3-VL-8B-Instruct [30]	-	38.51	40.75	39.63	39.63
InternVL3-8B [31]	-	37.54	41.85	40.50	39.97
GPT-4o [32] [†]	-	39.48	40.57	41.45	40.50
CLIP ViT-L/14 [33]	ICML 2021	47.40	56.42	56.41	53.41
FDIL (UTD only)	Ours	50.20 $\pm$ 0.44	58.03 $\pm$ 0.11	58.79 $\pm$ 0.78	55.68 $\pm$ 0.44
FDIL	Ours	52.13 $\pm$ 0.81	60.06 $\pm$ 0.24	60.28 $\pm$ 0.42	57.49 $\pm$ 0.49

with frozen CLIP text encoder [33]; their normalized logits are averaged before local reranking within **Top-10** visual candidates. For UTD, rationale text is generated offline by Qwen2.5-VL-7B-Instruct [35] and then encoded by a frozen bge-large-en-v1.5 text encoder [36]. **The exact scenario- and rationale-generation prompt templates are released together with our artifacts.** The teacher consumes only the resulting text embeddings and is optimized with Asymmetric Loss (ASL) [17] using $\gamma_- = 2$, $\gamma_+ = 0$, and clipping value 0.05. The student is optimized with AdamW [37], learning rate 5×10^{-4} , weight decay 10^{-4} , batch size 256, and dropout 0.1. The teacher’s hidden dimension is 1,024, and the student’s hidden dimension is 768. The distillation temperature is set to 2.0. **Best-epoch selection and early stopping use validation class-wise Macro F1 only. All decision thresholds are fitted exclusively on the validation split and then fixed before test evaluation; test outputs are used solely for final reporting and never enter any model- or threshold-selection branch.** Notably, UTD introduces *zero extra inference cost*, since the text teacher is only used during training and is discarded at test time.

4.2. Comparative Experiments

4.2.1. Comparison with state-of-the-art methods

Table 2 summarizes the main comparison on the test split. **The strong CLIP baseline already surpasses the previously reported best result by a clear margin; however, because the previous methods are quoted under their original, generally weaker backbones, this margin mainly reflects backbone strength rather than algorithmic superiority, and we therefore do not interpret it as evidence for FDIL.** For a fair comparison, we additionally reproduce the core mechanism of representative recent methods over the same frozen CLIP ViT-L/14 features, split, and validation-only class-wise thresholding, shown as the “(CLIP ViT-L/14)” rows. Under this matched protocol FDIL still leads all of them by +3.4 to +4.7 average F1. Building on the same features, UTD alone improves the average F1 over the baseline at zero extra inference cost, and the full FDIL framework further improves all four metrics. This pattern is consistent with our motivation that fixed semantic priors supply structured candidate semantics while

uncertainty-aware supervision and image-aware refinement sharpen the final prediction.

4.2.2. Zero-shot Multimodal Large Model Comparison

To address the concern that recent multimodal large language models may already solve visual intent recognition through prompting alone, we evaluate two recent open-weight MLLMs zero-shot on the same Intentionomy label set, prompting each to return a probability over the candidate categories; the group additionally lists, for reference, the GPT-4o zero-shot result reported by IntentMLM [10]. As shown in the *Zero-shot VLM* group of Table 2, direct zero-shot prediction is competitive but remains below the trained CLIP and FDIL models. Stronger general-purpose MLLMs therefore provide useful intent priors, but prompt-only recognition does not remove the need for dataset-specific calibration, uncertainty-aware supervision, and image-conditioned refinement.

4.2.3. Performance on Difficulty- and Content-Dependent Subsets

To better understand the effectiveness of our method, we further evaluate performance on intent subsets grouped by difficulty (Table 3) and by content dependency (Table 4). The difficulty-wise results provide two additional insights. First, the CLIP baseline already matches or exceeds prior methods across the three difficulty groups; on the easy subset FDIL is the stronger of our two models and trails only HLEG by 0.30, a negligible gap on visually unambiguous categories where modelling semantic overlap offers little headroom. Second, FDIL yields its largest and most consistent gains on the medium and hard subsets, precisely where semantic ambiguity and supervisory uncertainty are most pronounced and which the framework is designed to target.

Turning to content dependency, Table 4 reports results on content-dependent subsets. It again shows that the strong CLIP baseline is already highly effective on object- and context-dependent intents. The major gain of FDIL appears in the *Others* subset, where intent recognition requires more holistic reasoning over multiple weak cues rather than direct reliance on a single object or scene pattern.

Table 3: Difficulty-based intent subsets (Easy/Medium/Hard); Avg. is the subset mean. FDIL is the mean over three seeds.

Method	Easy	Medium	Hard	Avg.
Random	19.86	7.11	2.81	9.93
Visual [21]	54.64	24.92	10.71	30.09
Jia (w/o hashtag) [1]	57.10	25.68	12.72	31.83
Jia (hashtag) [1]	58.86	26.30	13.11	32.76
Query2Label [25]	81.59	38.06	14.55	44.73
CPAD [6]	69.47	31.50	8.54	36.50
HLEG [7]	81.49	38.99	20.22	46.90
PIP-Net [5]	70.73	33.63	14.33	39.56
IntentMLM [10]	76.26	46.87	27.39	50.17
CLIP ViT-L/14 [33]	80.99	53.32	28.45	54.25
FDIL	81.19	57.73	35.02	57.98

Table 4: Content-dependent intent subsets (Object/Context/Others); Avg. is the subset mean. FDIL is the mean over three seeds.

Method	Object	Context	Others	Avg.
Random	7.75	12.53	6.05	8.78
Visual [21]	25.58	30.16	21.34	25.69
Jia (w/o hashtag) [1]	28.15	28.62	22.60	26.46
Jia (hashtag) [1]	29.66	32.48	22.61	28.25
CPAD [6]	28.81	47.30	24.74	33.62
PIP-Net [5]	39.20	39.53	27.59	35.44
IntentMLM [10]	50.13	49.69	39.75	46.52
CLIP ViT-L/14 [33]	61.80	61.69	40.59	54.69
FDIL	63.04	61.32	47.15	57.17

4.2.4. Interpretability Analysis

To make the source of the improvement interpretable rather than aggregate, we examine where in the label space FDIL helps. Figure 3 shows a t-SNE projection of the decision-score space on four semantically adjacent intents: FDIL produces more compact, better-separated same-intent clusters than the CLIP baseline (silhouette 0.21 $\rightarrow$ 0.29), indicating that the semantic prior and the rationale teacher together pull confusable neighboring intents apart rather than merely rescaling scores. We further compute the per-class F1 change of FDIL relative to the CLIP baseline under the matched class-wise protocol. The largest per-class gains are concentrated on hard and abstract classes, including *WorkILike* (+28.5), *PassionAboutSomething* (+17.3), and *SocialLifeFriendship* (+17.2), while only two classes regress.

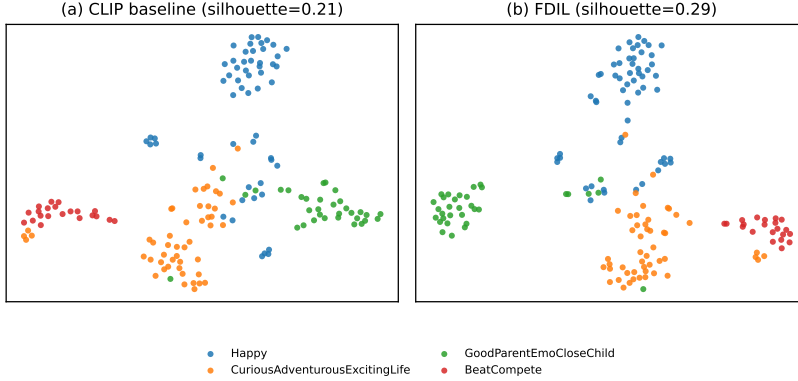


Figure 3: t-SNE of the decision-score space on four semantically adjacent intents. FDIL yields tighter, better-separated same-intent clusters than the CLIP baseline.

4.3. Ablation Studies

4.3.1. Effectiveness of SLR-C and Residual Refinement

We first study how SLR-C interacts with residual learning and distillation. Table 5 shows that prior alone is insufficient, while adding a trainable student and distillation is essential for recovering robust intent predictions.

Two observations are noteworthy. First, naively injecting a heterogeneous prior is not enough; without a trainable student, the mixed prior can distort the score geometry

Table 5: Component ablation of FDIL (*w/o UTD*: no text distillation; *w/o Gated*: no agreement-aware gating). Full FDIL is the mean over three seeds; ablations use a single seed.

Method	Macro	Micro	Samples	Avg.	Hard
w/o UTD	47.90	55.98	55.18	53.02	29.71
w/o Gated	49.69	57.90	58.98	55.52	31.69
FDIL (full)	52.13	60.06	60.28	57.49	35.02

and lead to poor predictions. Second, once distillation is introduced, residual refinement becomes consistently beneficial, showing that heterogeneous prior fusion and uncertainty-aware supervision must be optimized jointly rather than independently.

4.3.2. Unified versus Functionally Decoupled Ambiguity Handling

We next examine whether the functional decomposition is necessary, or whether a single unified mechanism can absorb both forms of ambiguity. Table 6 contrasts the decoupled FDIL model with three unified controls under identical frozen CLIP ViT-L/14 features, data split, and validation-only class-wise thresholding. The most direct control, *Unified joint-target KD*, averages the SLR-C and rationale-teacher probabilities into a single distillation target and trains one image-feature student, without residual decomposition or agreement-aware gating.

Table 6: Unified versus functionally decoupled ambiguity handling. Results are means over three seeds.

Method	Macro	Micro	Samples	Avg.	Hard
Unified UTD only	50.20	58.03	58.79	55.68	30.27
Unified SLR-C only	51.29	60.10	59.22	56.87	34.01
Unified joint-target KD	50.69	58.35	57.96	55.66	32.41
FDIL (decoupled)	52.13	60.06	60.28	57.49	35.02

The results show that neither signal is degenerate in isolation: the semantic prior alone and the rationale teacher alone are each competitive. Crucially, the manner of combination is decisive. Collapsing the two signals into a single mixed target (*Unified joint-target KD*) yields no benefit, whereas the decoupled FDIL model is best overall.

We therefore interpret decoupling as conferring a small but consistent accuracy gain over a single mixed target, together with its primary advantage of modularity: each function is separately justified and measurable, yielding an independent mAP increment, and the UTD branch can be deployed alone at zero additional inference cost.

4.3.3. Heterogeneous Priors in SLR-C

We next compare different textual priors used by SLR-C. We consider three semantic sources: *Lexical*, which uses short class-name phrases; *Canonical*, which uses the official intent descriptions provided by Intentionomy; and *Scenario*, which uses Gemini-generated visual scenarios. For the ensemble setting, we average the normalized logits from the selected sources before local reranking. Table 7 therefore isolates the effect of prior sources under the SLR-C reranking stage only.

Table 7: Ablation of textual priors in SLR-C (reranking-only, $K=10$). Results use a single seed.

Lexical	Canonical	Scenario	Macro	Micro	Samples	Avg.	Hard
✓			49.76	59.12	58.60	55.83	31.22
	✓		49.80	59.50	58.05	55.78	29.88
		✓	49.35	58.88	57.87	55.37	32.48
✓	✓		51.18	59.39	58.11	56.23	32.92
✓		✓	50.12	59.30	58.36	55.93	32.75
	✓	✓	50.93	59.51	58.18	56.21	34.81
✓	✓	✓	50.90	59.81	58.47	56.40	32.80

The results show that ensemble outperforms its single-source components on the aggregate metrics, confirming that the three sources carry complementary information. Lexical phrases are compact, canonical descriptions are semantically explicit, and scenario descriptions ground latent intention into typical visual situations. Second, no single source dominates at the reranking-only stage, the choice of prior is a trade-off rather than a strict ordering. Third, the three-source ensemble achieves the best aggregate performance and the broadest semantic coverage, and once combined with residual refinement and the UTD teacher it also yields the strongest hard-subset result.

We therefore adopt it in FDIL for its balanced coverage and strong behaviour under the full pipeline.

4.3.4. Sensitivity to Candidate Size K

We next study the effect of candidate size K while fixing the thresholding strategy to the final class-wise setting. Table 8 reports the test-split results obtained by retraining FDIL under $K \in \{5, 10, 28\}$ and then searching thresholds on the validation split. The comparison shows that the best overall result is achieved at $K = 10$, which we adopt as the default candidate size; $K = 5$ is slightly weaker and $K = 28$ is comparable but offers no further benefit. This suggests that, under the final lexical-canonical-scenario prior, a moderate local candidate set best balances semantic coverage against the distractor noise introduced by an overly broad reranking range.

Table 8: Sensitivity to candidate size K (class-wise thresholding). Results are means over three seeds.

Top- K	Macro	Micro	Samples	Avg.	Hard
5	51.19	59.03	59.42	56.54	31.60
10	52.13	60.06	60.28	57.49	35.02
28	51.64	59.59	59.71	56.98	34.02

Table 9: Calibration decomposition under matched global/class-wise thresholding. FDIL is mean $\pm$ std over three seeds; CLIP uses a single seed.

Method	Threshold	Macro	Micro	Samples	Avg.	mAP	Hard
CLIP ViT-L/14 [33]	Global	46.22	55.05	54.51	51.92	50.67	26.84
CLIP ViT-L/14 [33]	Class-wise	47.40	56.42	56.41	53.41	50.67	28.45
FDIL	Global	47.70 $\pm$ 1.55	57.56 $\pm$ 0.41	57.43 $\pm$ 0.69	54.23 $\pm$ 0.72	55.10 $\pm$ 0.26	30.00 $\pm$ 3.46
FDIL	Class-wise	52.13 $\pm$ 0.81	60.06 $\pm$ 0.24	60.28 $\pm$ 0.42	57.49 $\pm$ 0.49	55.10 $\pm$ 0.26	35.02 $\pm$ 1.77

4.3.5. Calibration and Threshold-free Evaluation

To separate FDIL’s mechanisms from class-wise calibration, we evaluate both the CLIP baseline and FDIL under matched global and class-wise thresholding (Table 9). FDIL improves over the baseline under both protocols on average and hard-subset

F1, and class-wise thresholding helps both models, so calibration is a shared post-processing gain rather than the source of FDIL’s advantage. The threshold-free mAP, invariant to the decision rule by construction, confirms the same conclusion with a +4.4 gain.

Table 10: Threshold-free, rank-based metrics. mAP is the macro average precision. FDIL/UTD mAP values are means over three seeds; all AUCs and CLIP mAP use a single representative seed.

Method	mAP	Macro ROC-AUC	Micro ROC-AUC	Micro PR-AUC
CLIP ViT-L/14 [33]	50.67	87.04	84.90	54.89
UTD only	53.07	90.09	88.55	59.59
SLR-C only	53.79	88.24	86.73	58.60
FDIL	55.10	89.36	88.31	60.64

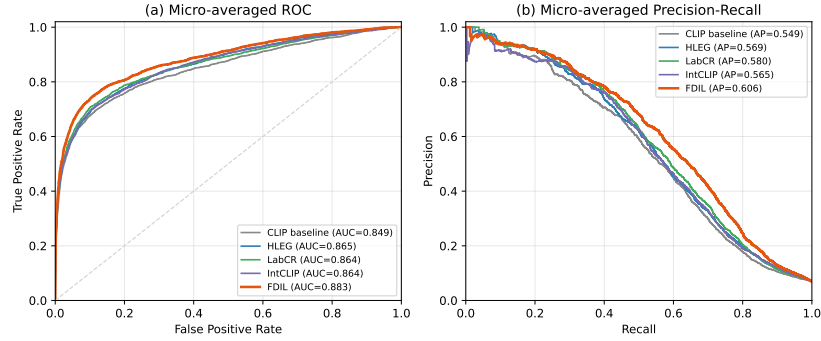


Figure 4: Micro-averaged ROC (a) and precision-recall (b) curves, comparing FDIL against the CLIP baseline and three representative methods (HLEG, LabCR, IntCLIP) reproduced under the matched CLIP ViT-L/14 features, split, and protocol.

Table 10 and Figure 4 report fully threshold-free, rank-based metrics (mAP, ROC-AUC, PR-AUC) for FDIL, its ablations, and representative methods reproduced under the matched CLIP ViT-L/14 protocol. FDIL improves over the baseline on every metric with a 95% paired-bootstrap interval excluding zero ($p < 0.001$), and FDIL outperforms the reproduced methods when the backbone is equalized.

4.3.6. The Effectiveness of Rationale

We then compare generic image captioning with rationale-style supervision under UTD only setting. We further ablate the internal structure of the rationale used by the text teacher. Specifically, we compare Caption, which only describes visible content in short natural language; Step 1 only, which contains observable visual evidence; Step 1 with Step 2, which additionally introduces contextual interpretation; and Full rationale, which further adds counterfactual disambiguation cues. Table 11 reports the results under gated distillation without SLR-C.

Table 11: Ablation of rationale construction for UTD (Step 1/2/3: visual evidence, contextual bridging, counterfactual disambiguation). Results use a single seed.

Method	Macro	Micro	Samples	Avg.	Hard
Caption	49.64	58.64	58.29	55.52	29.14
Rationale (Step 1)	49.81	56.39	57.65	54.62	29.28
Rationale (Step 1 & 2)	49.69	54.45	55.97	53.37	28.82
Rationale (Step 1 & 2 & 3)	50.31	57.97	59.07	55.78	30.24

The trend is revealing. Caption supervision is already a reasonably strong privileged signal, but it remains slightly weaker than the full rationale in Macro, Samples, Avg., and Hard. Step 1 already provides useful visual facts, but adding Step 2 alone slightly degrades the overall classification metrics, suggesting that contextual interpretation without explicit disambiguation may introduce semantic drift. Once the counterfactual Step 3 is added, however, the rationale becomes significantly more effective. This supports our central claim that subjective intention recognition requires not only richer context, but also explicit boundary contraction among confusing alternatives.

4.3.7. UTD Ablations: Gating and Teacher Modality

We isolate the effect of UTD without using the SLR-C prior, jointly studying agreement-aware gating and the teacher’s input modality. Table 12 (top block) compares plain supervised learning, distillation without gating, and agreement-aware gated distillation; the bottom block contrasts teacher-only oracle performance with the cor-

responding gated-distillation student under text-only versus image+text teacher inputs.

Table 12: UTD teacher ablations without SLR-C. Top: agreement-aware gating. Bottom: teacher input modality (oracle and gated student). Results use a single seed; bold marks the best entry within each comparison.

Method	Modality	Macro	Micro	Samples	Avg.	Hard
<i>Effect of agreement-aware gating</i>						
Baseline	–	47.40	56.42	56.41	53.41	28.45
UTD (w/o gated)	Text	49.59	58.26	58.88	55.58	30.44
UTD (with gated)	Text	50.31	57.97	59.07	55.78	30.24
<i>Effect of teacher input modality</i>						
Oracle teacher	Text	45.80	51.42	52.61	49.94	30.21
Oracle teacher	Image+Text	45.87	53.44	54.04	51.12	28.08
Gated student	Text	50.31	57.97	59.07	55.78	30.24
Gated student	Image+Text	48.57	56.10	57.13	53.93	28.58

Table 12 shows that both ungated and gated UTD improve over the supervised baseline, while agreement-aware gating gives the best Avg. F1. The modality comparison further shows that a multimodal oracle teacher is not weaker by itself, but distills worse than the text-only teacher, suggesting that the text manifold provides cleaner semantic supervision.

4.3.8. Negative Controls on the Text Teacher

A natural concern is whether UTD’s gains arise from any external VLM text added on top of a strong backbone, rather than from the structured, uncertainty-gated rationale specifically. We test this with negative-control retrains, each corrupting a single ingredient of the pipeline while keeping the rest fixed (Table 13): shuffled rationale, where each image is paired with another image’s rationale so the teacher carries no per-sample information; ungated teacher, where the agreement gate is removed (standard KD); and uniform gate, where every sample receives the mean agreement. The supervised residual without distillation (w/o UTD) serves as the lower reference.

Table 13: Negative controls on the UTD text teacher (Δ in points; negative denotes degradation). Full FDIL is the mean over three seeds; control deltas use a single seed.

Condition	Macro	Micro	Samples	Avg.	mAP	Hard
FDIL (full)	52.13	60.06	60.28	57.49	55.10	35.02
w/o UTD (supervised)	-4.03	-5.59	-5.17	-4.94	-2.18	-6.02
Shuffled rationale	-7.75	-12.39	-11.01	-10.40	-9.04	-7.58
Ungated teacher	-0.62	-0.41	+0.07	-0.33	+0.03	-0.69
Uniform agreement gate	-1.16	-0.93	-1.21	-1.12	-1.33	+1.13

Table 13 verifies that UTD does not benefit from arbitrary external text. Shuffling rationales causes a larger degradation than removing UTD, confirming that the gain relies on image-specific rationale content. Flattening or removing the agreement gate leads to smaller but consistent degradation, supporting the use of uncertainty-aware gating.

4.4. Backbone Independence

To examine whether FDIL depends entirely on a strong backbone, we also test the full pipeline with a weaker ResNet101 student under the best configuration. The results are shown in Table 14. The value of K is set to 28 to give the weaker backbone the widest semantic candidate scaffold.

Table 14: Backbone sensitivity of FDIL on the Intentionomy test split. CLIP FDIL is the mean over three seeds; ResNet101 results use a single seed.

Backbone	Method	Macro	Micro	Samples	Avg.	Hard
ResNet101 [21]	Baseline	36.56	41.87	43.10	40.51	18.99
ResNet101 [21]	IntCLIP [9]	40.15	53.54	51.01	48.23	-
ResNet101 [21]	FDIL	43.20	50.63	52.02	48.62	25.65
CLIP ViT-L/14 [33]	Baseline	47.40	56.42	56.41	53.41	28.45
CLIP ViT-L/14 [33]	FDIL	52.13	60.06	60.28	57.49	35.02

Three conclusions follow from Table 14. First, FDIL remains clearly effective on ResNet101 and shows that the functionally decoupled design is not exclusive to CLIP

features. Second, compared with IntCLIP, which achieves state-of-the-art under the same ResNet101 backbone, FDIL achieves **a better result**. Third, the best ResNet101 configuration prefers a much larger candidate set than the CLIP backbone, suggesting that weaker visual representations benefit from a broader semantic scaffold before residual correction. At the same time, the remaining gap to CLIP ViT-L/14 indicates that strong visual features are still important for fully exploiting the semantic prior and the uncertainty-aware teacher.

4.5. Efficiency Comparison

Finally, we compare the inference cost of the baseline, FDIL (UTD only), and the full FDIL pipeline, **together with the representative external methods**. Since all these methods share the same frozen CLIP ViT-L/14 feature extractor, Table 15 reports the additional inference cost on top of pre-extracted CLIP features. Parameters count the inference-active prediction heads only, FLOPs are computed per image under the same feature-level protocol, and latency is measured with batch size 1 on a single RTX 4090 GPU.

Table 15: **Feature-level inference cost on frozen CLIP ViT-L/14 features: params (inference-active heads), per-image FLOPs, and batch-1 latency on an RTX 4090.**

Method	Params (M)	FLOPs (M)	Latency (ms)
Baseline	1.20	2.40	0.109
FDIL (UTD only)	1.20	2.40	0.109
FDIL	2.41	4.93	0.466
HLEG [7]	203.70	12 075.9	2.725
IntCLIP [9]	0.02	22.02	0.111
IntentMLM [10]	MLLM: $\sim 10^9$ params, 10–100 ms/image		

The table confirms two intended properties of our design. First, FDIL (UTD only) has exactly the same inference cost as the baseline, because the text teacher is used only during training and discarded at test time. Second, the full FDIL pipeline introduces additional reranking and residual refinement, but the absolute overhead remains small:

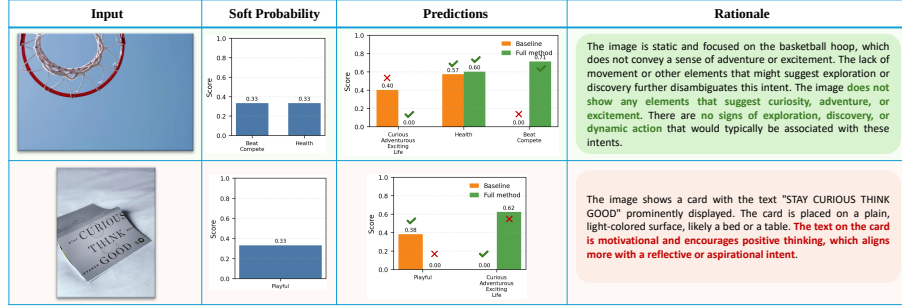


Figure 5: Qualitative case study of baseline and FDIL on representative Intentionomy samples.

even with the full lexical-canonical-scenario prior and residual head, the feature-level inference latency stays well below 1 ms per image. This shows that the final performance gains are not obtained by paying a large deployment cost.

Profiling the previous methods under the same shared-feature protocol places this overhead in context. HLEG’s decoder costs two-to-three orders of magnitude more than FDIL, whereas IntCLIP uses a tiny learnable head whose region aggregation still incurs non-trivial FLOPs. FDIL sits between these regimes, keeping sub-millisecond latency while adding an explicit semantic prior and residual refinement; IntentMLM is dominated by an MLLM and reported only qualitatively.

4.6. Case Study

Figure 5 provides qualitative examples. In the first case, FDIL suppresses a plausible but incorrect distractor by using counterfactual rationale evidence, consistent with the role of UTD in sharpening semantic boundaries. In the second case, the annotated label has only 0.33 agreement, and the FDIL prediction remains semantically plausible, illustrating the intrinsic uncertainty of subjective intent recognition.

5. Discussion

5.1. A Functionally Decoupled View of Ambiguity

The results support our premise that visual-intent ambiguity is functionally decoupled: supervisory ambiguity destabilizes the training target, whereas semantic ambiguity distorts the local decision geometry among neighboring intents. This explains

why combining the semantic prior and rationale teacher into a single target is weaker than FDIL, while the decoupled formulation keeps each function separately justified, measurable, and deployable.

This viewpoint also clarifies why a strong CLIP baseline already performs well. Once visual representation is sufficiently expressive, the remaining bottleneck is no longer basic visual recognition, but the mapping from rich visual semantics to abstract intent labels. In other words, the main challenge is not merely seeing better, but deciding better under ambiguity.

5.2. Limitations and Future Directions

Despite the consistent gains, the framework has several limitations. First, it depends on offline text generation and external prior construction, increasing preparation cost, even though UTD adds no inference-time cost. Second, the gains shrink on weaker backbones such as ResNet101, indicating that residual refinement still relies on reasonably strong base representations. Third, richer priors are not automatically better unless their interaction with the student is carefully controlled. Fourth, per-class attribution reveals a failure mode: for a few broadly defined, overlapping categories, the residual student can resolve the local candidate set toward a competing neighbor and lose accuracy on the original class. Fifth, we add a transfer analysis on EMOTIC [38], a related affective understanding dataset. Under a five-seed, five-fold protocol, FDIL improves over the baseline on Micro F1, Samples F1, and mAP, with significant gains on Micro F1 and mAP. Ablations show that the improvement mainly comes from agreement-gated UTD, while SLR-C is more dependent on the compatibility between label semantics and the target label space.

These limitations suggest several future directions: learning source-aware prior weighting instead of uniform averaging, designing stronger students for weaker backbones, and incorporating explicit candidate-level reasoning so that textual guidance operates through confusion-pair-specific evidence resolution rather than only global distillation and local reranking.

6. Conclusion

In this work, we study subjective visual intention recognition under supervisory ambiguity and semantic ambiguity, and propose FDIL as a functionally decoupled framework. In the current formulation, fixed semantic priors provide structured candidate semantics, a residual student supports image-aware intent prediction on top of the semantic prior, and uncertainty-aware text teachers stabilize ambiguous supervision.

Extensive experiments on Intentionomy show that the proposed framework consistently improves over strong visual baselines and prior state-of-the-art methods. Overall, the results suggest that once visual representations are sufficiently strong, the key challenge of intention understanding lies in resolving supervisory and semantic ambiguity rather than in extracting low-level visual features alone. We hope this functionally decoupled formulation provides a clearer perspective for future research on human-centered visual understanding tasks.

Data and artifact availability

The Intentionomy benchmark is publicly available. Upon acceptance, we will release generated priors, rationales, cached features/logits, prompts, thresholds, seeds, checksums, and training/evaluation scripts.

CRedit authorship contribution statement

Yin Tang: Writing - original draft, Validation, Formal analysis; **Lifeng Fan:** Supervision, Writing - original draft, Conceptualization; **Hongyu Yang:** Writing - original draft, Formal analysis; **Xuan Dong:** Writing - original draft, Formal analysis; **Weixin Li:** Writing - original draft, Supervision, Resources, Project administration.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used ChatGPT in order to improve language and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

- [1] Menglin Jia, Zuxuan Wu, Austin Reiter, Claire Cardie, Serge Belongie, and Ser-Nam Lim. Intentionomy: a Dataset and Study towards Human Intent Understanding. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12981–12991, June 2021. ISSN: 2575-7075.
- [2] Changsong Wen, Guoli Jia, and Jufeng Yang. DIP: Dual Incongruity Perceiving Network for Sarcasm Detection. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2540–2550, Vancouver, BC, Canada, June 2023. IEEE.
- [3] Joshua Peterson, Ruairidh Battleday, Thomas Griffiths, and Olga Russakovsky. Human Uncertainty Makes Classification More Robust. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9616–9625, Los Alamitos, CA, USA, November 2019. IEEE Computer Society.
- [4] Aida Mostafazadeh Davani, Mark Díaz, and Vinodkumar Prabhakaran. Dealing with Disagreements: Looking Beyond the Majority Vote in Subjective Annotations. *Transactions of the Association for Computational Linguistics*, 10:92–110, 2022.
- [5] Binglu Wang, Kang Yang, Yongqiang Zhao, Teng Long, and Xuelong Li. Prototype-Based Intent Perception. *IEEE Transactions on Multimedia*, 25:8308–8319, 2023.
- [6] Mang Ye, Qinghongya Shi, Kehua Su, and Bo Du. Cross-Modality Pyramid Alignment for Visual Intention Understanding. *IEEE Transactions on Image Processing*, 32:2190–2201, 2023.
- [7] QingHongYa Shi, Mang Ye, Ziyi Zhang, and Bo Du. Learnable Hierarchical Label Embedding and Grouping for Visual Intention Understanding. *IEEE Trans. Affect. Comput.*, 14(4):3218–3230, 2023.

- [8] QingHongYa Shi, Mang Ye, Wenke Huang, Weijian Ruan, and Bo Du. Label-Aware Calibration and Relation-Preserving in Visual Intention Understanding. *IEEE Transactions on Image Processing*, 33:2627–2638, 2024.
- [9] Qu Yang, Mang Ye, and Dacheng Tao. Synergy of Sight and Semantics: Visual Intention Understanding with CLIP. In Aleš Leonardis, Elisa Ricci, Stefan Roth, Olga Russakovsky, Torsten Sattler, and Gül Varol, editors, *Computer Vision – ECCV 2024*, volume 15069, pages 144–160. Springer Nature Switzerland, Cham, 2025. Series Title: Lecture Notes in Computer Science.
- [10] Zhaozhong Wang, Yao Tian, Shutong Wang, Le Yang, and Binglu Wang. Unlocking human intent perception through multimodal large models. *Pattern Recognition*, 172:112582, April 2026.
- [11] Jungseock Joo, Weixin Li, Francis F. Steen, and Song-Chun Zhu. Visual Persuasion: Inferring Communicative Intents of Images. In *2014 IEEE Conference on Computer Vision and Pattern Recognition*, pages 216–223, June 2014. ISSN: 1063-6919.
- [12] Jungseock Joo, Francis F. Steen, and Song-Chun Zhu. Automated Facial Trait Judgment and Election Outcome Prediction: Social Dimensions of Face. In *2015 IEEE International Conference on Computer Vision (ICCV)*, pages 3712–3720, Santiago, Chile, December 2015. IEEE.
- [13] Xinyue Huang and Adriana Kovashka. Inferring Visual Persuasion via Body Language, Setting, and Deep Features. In *2016 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 778–784, Las Vegas, NV, USA, June 2016. IEEE.
- [14] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal Loss for Dense Object Detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 42(2):318–327, February 2020.
- [15] Abhinav Shrivastava, Abhinav Gupta, and Ross Girshick. Training Region-Based Object Detectors with Online Hard Example Mining. In *2016 IEEE Conference*

on *Computer Vision and Pattern Recognition (CVPR)*, pages 761–769, June 2016. ISSN: 1063-6919.

- [16] Mang Ye, He Li, Bo Du, Jianbing Shen, Ling Shao, and Steven C. H. Hoi. Collaborative Refining for Person Re-Identification With Label Noise. *IEEE Transactions on Image Processing*, 31:379–391, 2022.
- [17] Tal Ridnik, Emanuel Ben-Baruch, Nadav Zamir, Asaf Noy, Itamar Friedman, Matan Protter, and Lihi Zelnik-Manor. Asymmetric Loss For Multi-Label Classification. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 82–91, 2021.
- [18] Tharindu Cyril Weerasooriya, Alexander Ororbia, Raj Bhensadadia, Ashiqur KhudaBukhsh, and Christopher Homan. Disagreement Matters: Preserving Label Diversity by Jointly Modeling Item and Annotator Label Distributions with DisCo. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Findings of the Association for Computational Linguistics: ACL 2023*, pages 4679–4695, Toronto, Canada, July 2023. Association for Computational Linguistics.
- [19] Tharindu Cyril Weerasooriya, Sarah Luger, Saloni Poddar, Ashiqur KhudaBukhsh, and Christopher Homan. Subjective Crowd Disagreements for Subjective Data: Uncovering Meaningful CrowdOpinion with Population-level Learning. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 950–966, Toronto, Canada, July 2023. Association for Computational Linguistics.
- [20] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330. PMLR, July 2017.
- [21] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep Residual Learning for Image Recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.

- [22] Shu Zhang, Ran Xu, Caiming Xiong, and Chetan Ramaiah. Use All The Labels: A Hierarchical Multi-Label Contrastive Learning Framework. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16639–16648, June 2022. ISSN: 2575-7075.
- [23] Jinyuan Jia, Wenjie Qu, and Neil Gong. MultiGuard: Provably Robust Multi-label Classification against Adversarial Examples. *Advances in Neural Information Processing Systems*, 35:10150–10163, December 2022.
- [24] Tianshui Chen, Tao Pu, Hefeng Wu, Yuan Xie, and Liang Lin. Structured Semantic Transfer for Multi-Label Recognition with Partial Labels. *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(1):339–346, June 2022.
- [25] Shilong Liu, Lei Zhang, Xiao Yang, Hang Su, and Jun Zhu. Query2Label: A Simple Transformer Way to Multi-Label Classification. *arXiv preprint arXiv:2107.10834*, July 2021.
- [26] Ximeng Sun, Ping Hu, and Kate Saenko. DualCoOp: fast adaptation to multi-label recognition with limited annotations. In *Proceedings of the 36th International Conference on Neural Information Processing Systems, NIPS ’22*, pages 30569–30582, Red Hook, NY, USA, 2022. Curran Associates Inc.
- [27] Kuang-Huei Lee, Xiaodong He, Lei Zhang, and Linjun Yang. CleanNet: Transfer Learning for Scalable Image Classifier Training with Label Noise. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5447–5456, Salt Lake City, UT, June 2018. IEEE.
- [28] Michal Lukasik, Srinadh Bhojanapalli, Aditya Menon, and Sanjiv Kumar. Does label smoothing mitigate label noise? In *Proceedings of the 37th International Conference on Machine Learning*, pages 6448–6458. PMLR, November 2020.
- [29] Junnan Li, Richard Socher, and Steven C. H. Hoi. DivideMix: Learning with Noisy Labels as Semi-supervised Learning. In *International Conference on Learning Representations (ICLR)*, September 2020.

- [30] Shuai Bai, Yuxuan Cai, Ruizhe Chen, et al. Qwen3-vl technical report, 2025.
- [31] Jinguo Zhu, Weiyun Wang, Zhe Chen, et al. Internvl3: Exploring advanced training and test-time recipes for open-source multimodal models, 2025.
- [32] Aaron Hurst, Adam Lerer, Adam P. Goucher, Adam Perelman, Aditya Ramesh, et al. GPT-4o System Card, 2024.
- [33] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning*, pages 8748–8763. PMLR, July 2021.
- [34] Gemini Team. Gemini: A Family of Highly Capable Multimodal Models. *arXiv preprint arXiv:2312.11805*, May 2025.
- [35] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-VL Technical Report. *arXiv preprint arXiv:2502.13923*, February 2025.
- [36] Shitao Xiao, Zheng Liu, Peitian Zhang, Niklas Muennighoff, Defu Lian, and Jian-Yun Nie. C-Pack: Packed Resources For General Chinese Embeddings. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR '24*, pages 641–649, New York, NY, USA, 2024. Association for Computing Machinery.
- [37] Ilya Loshchilov and Frank Hutter. Decoupled Weight Decay Regularization. In *International Conference on Learning Representations (ICLR)*, September 2019.
- [38] Ronak Kosti, Jose M. Alvarez, Adria Recasens, and Agata Lapedriza. Context Based Emotion Recognition Using EMOTIC Dataset. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 42(11):2755–2766, 2019.